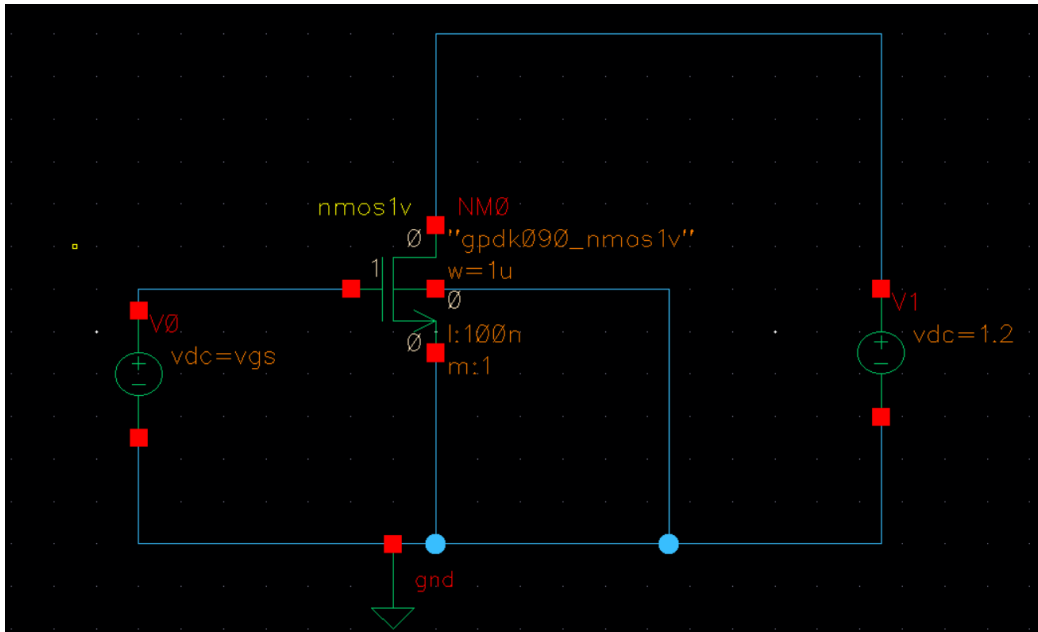


EXP 1: NMOS Characteristics:

Report:

The drain current (I_d) of an MOS transistor depends on the applied gate voltage (V_g).

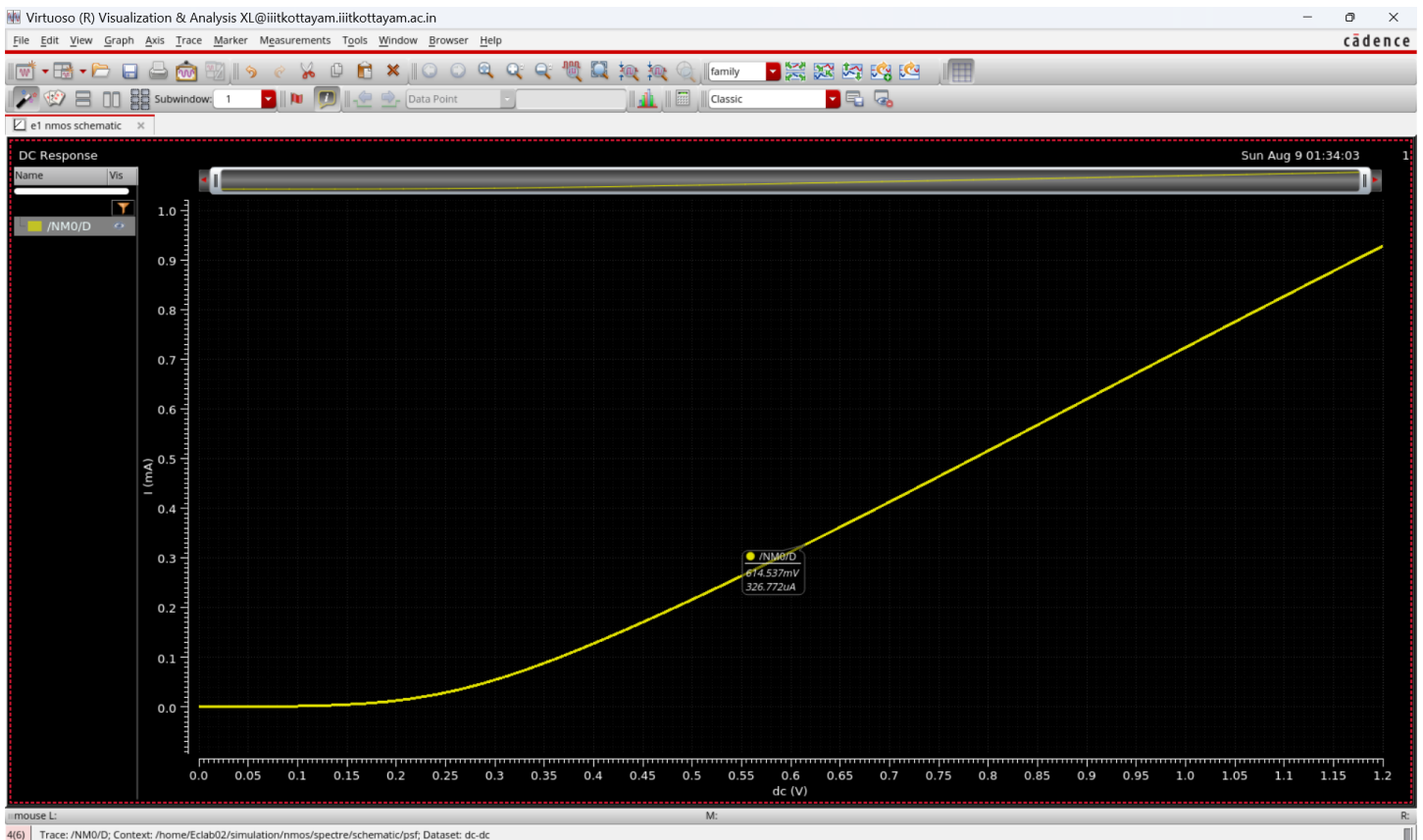
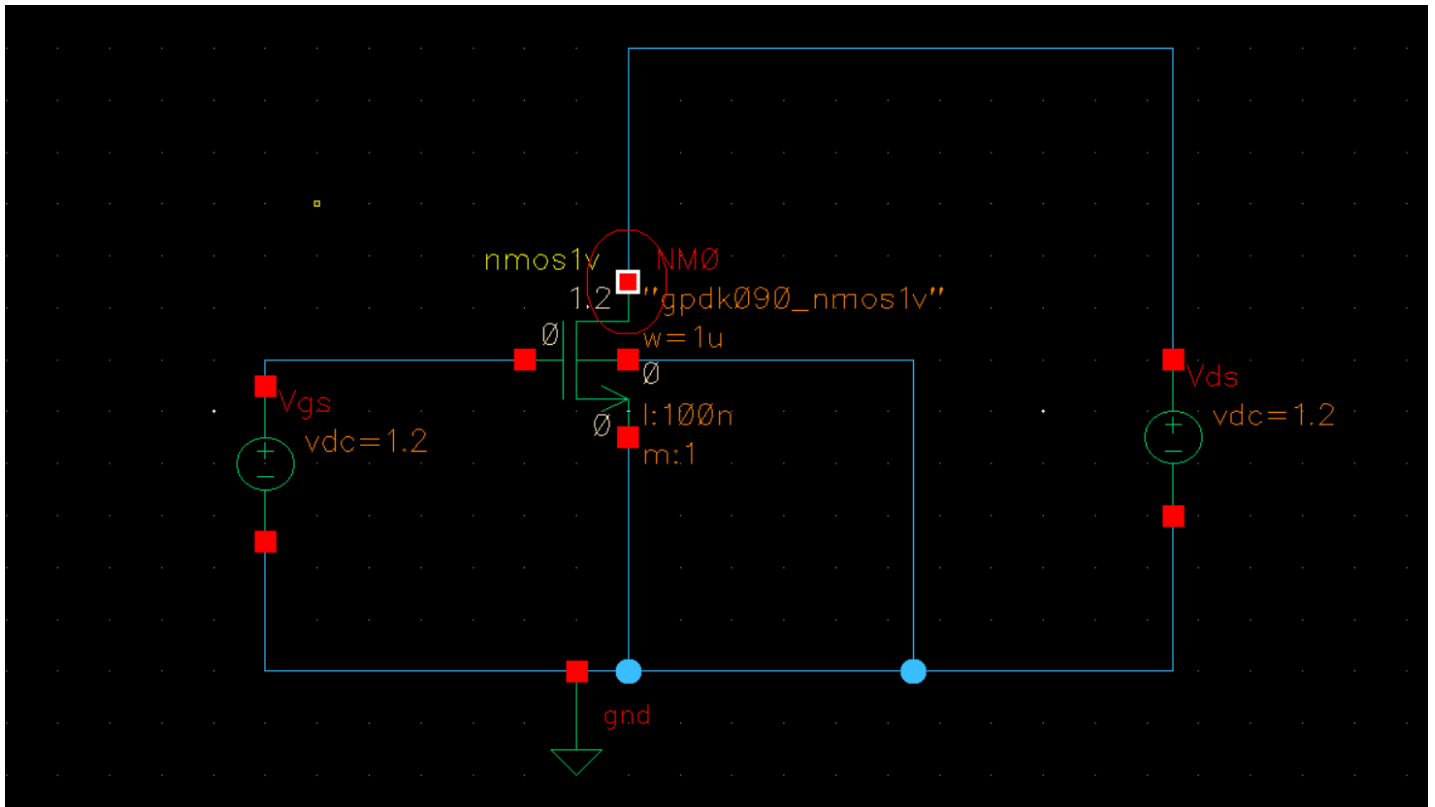
Schematic:



Simulations:

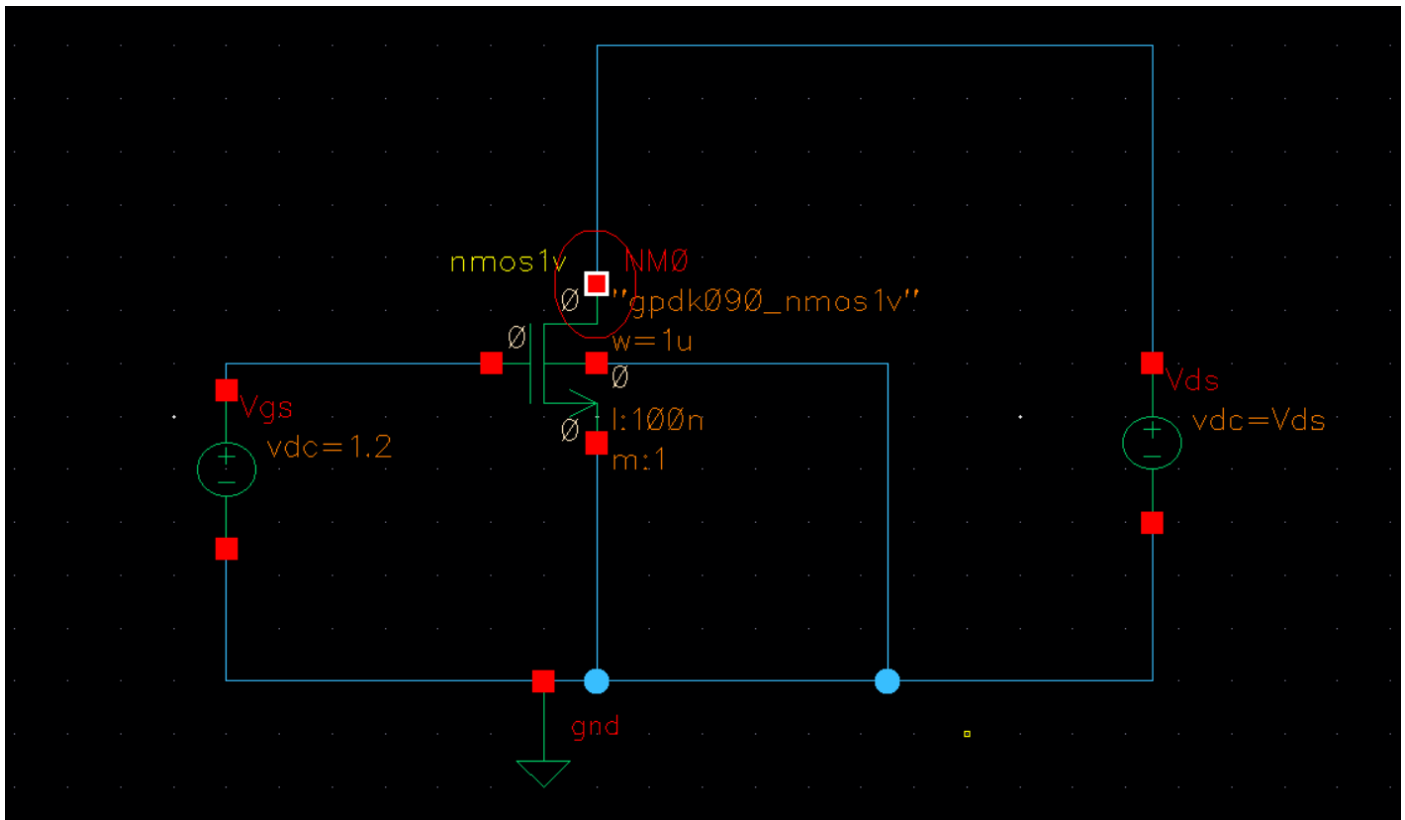
1. Plot I_d vs V_{gs} (V_{ds} is constant)

You can see $V_{ds} = 1.2\text{ V}$ (constant) in the schematic.



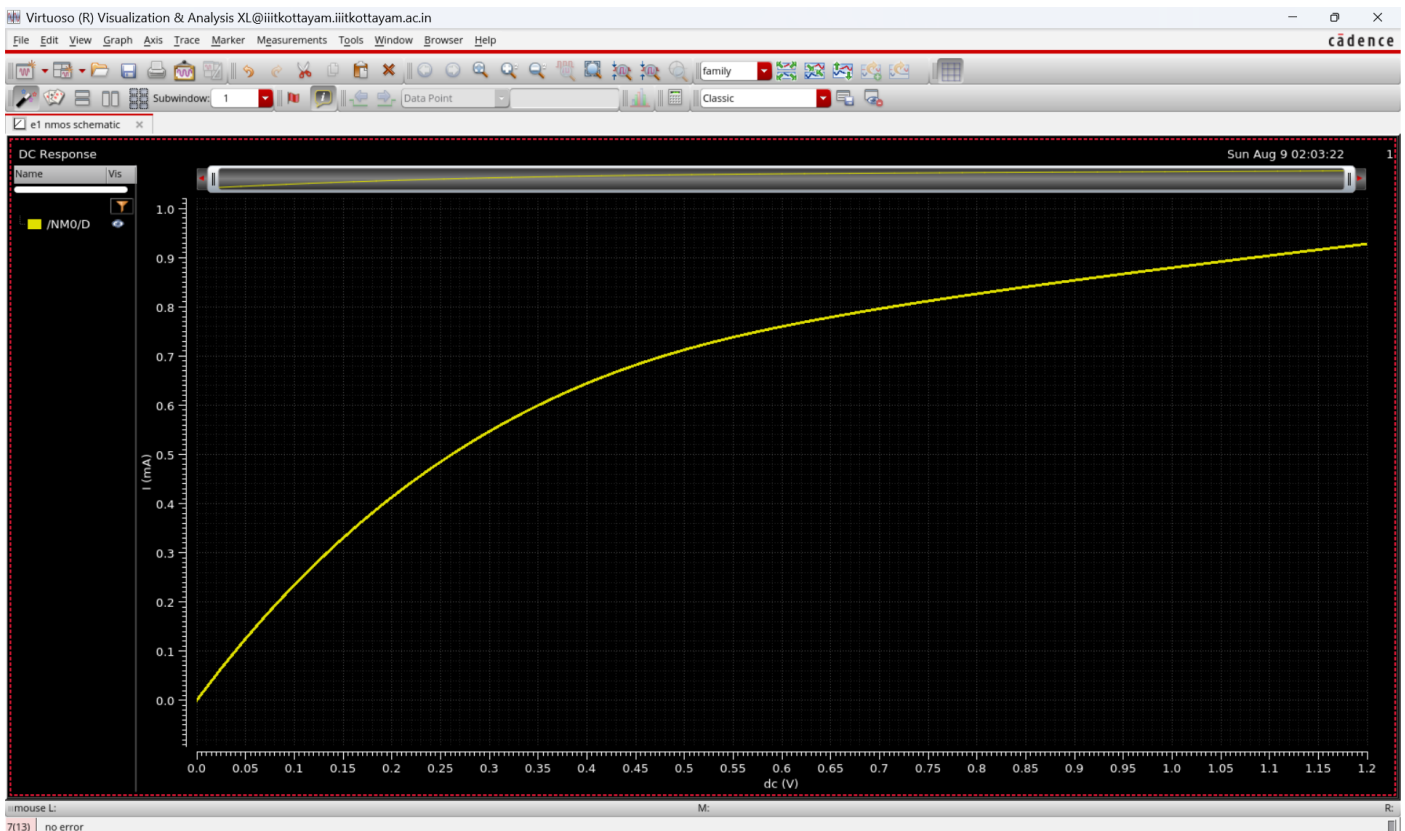
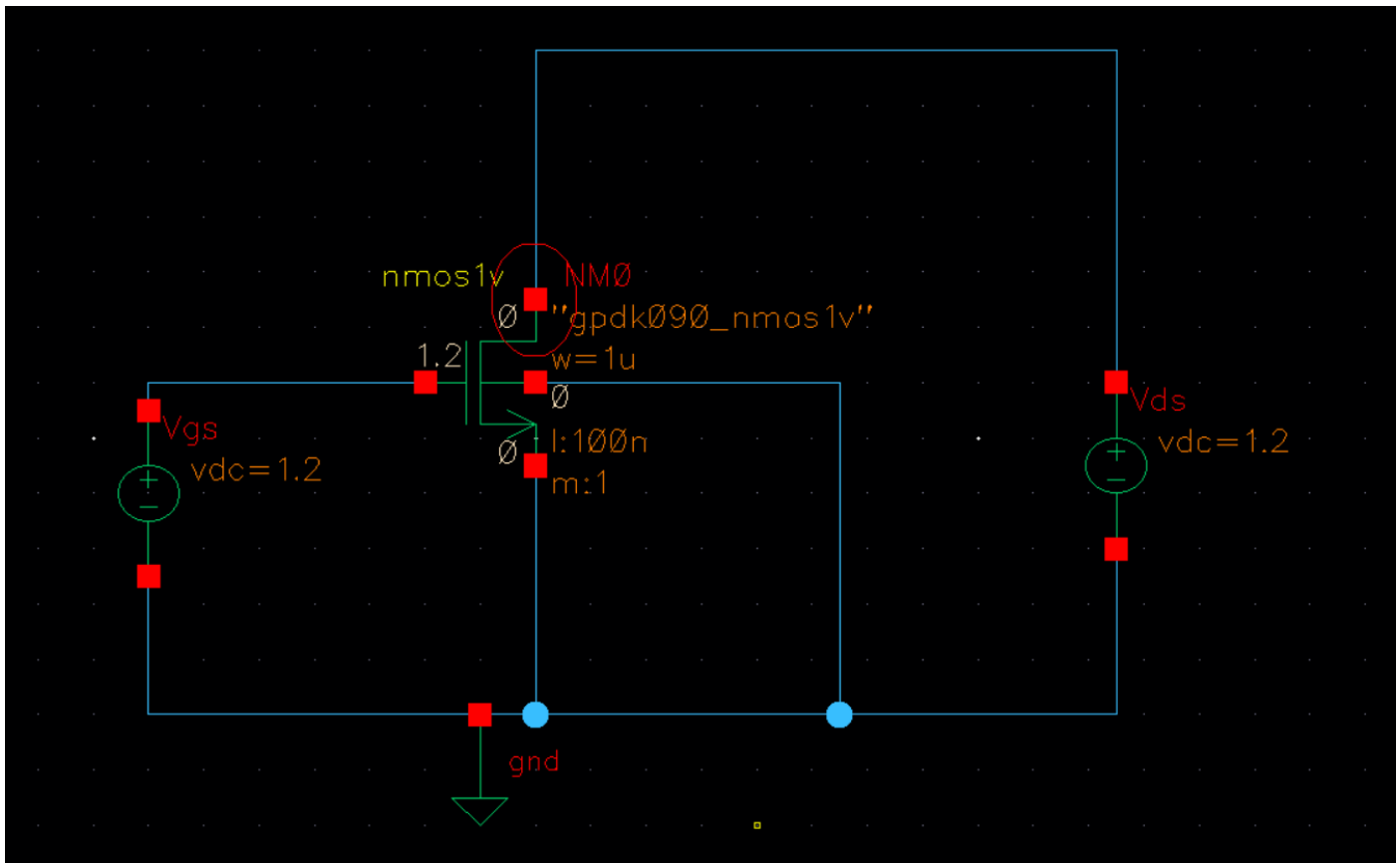
2) Plot I_d vs V_{gs} curve with different values of V_{ds} .

Now, you can see $V_{ds} = V_{ds}$ (parameter) in the schematic



3) Plot I_d vs V_{ds}

Respective schematic: Observe, $V_{gs} = 1.2\text{ V}$



4) Plot I_d vs V_{ds} (for different value of V_{gs})

$V_{gs} = V_{gs}$ (Parameter)

